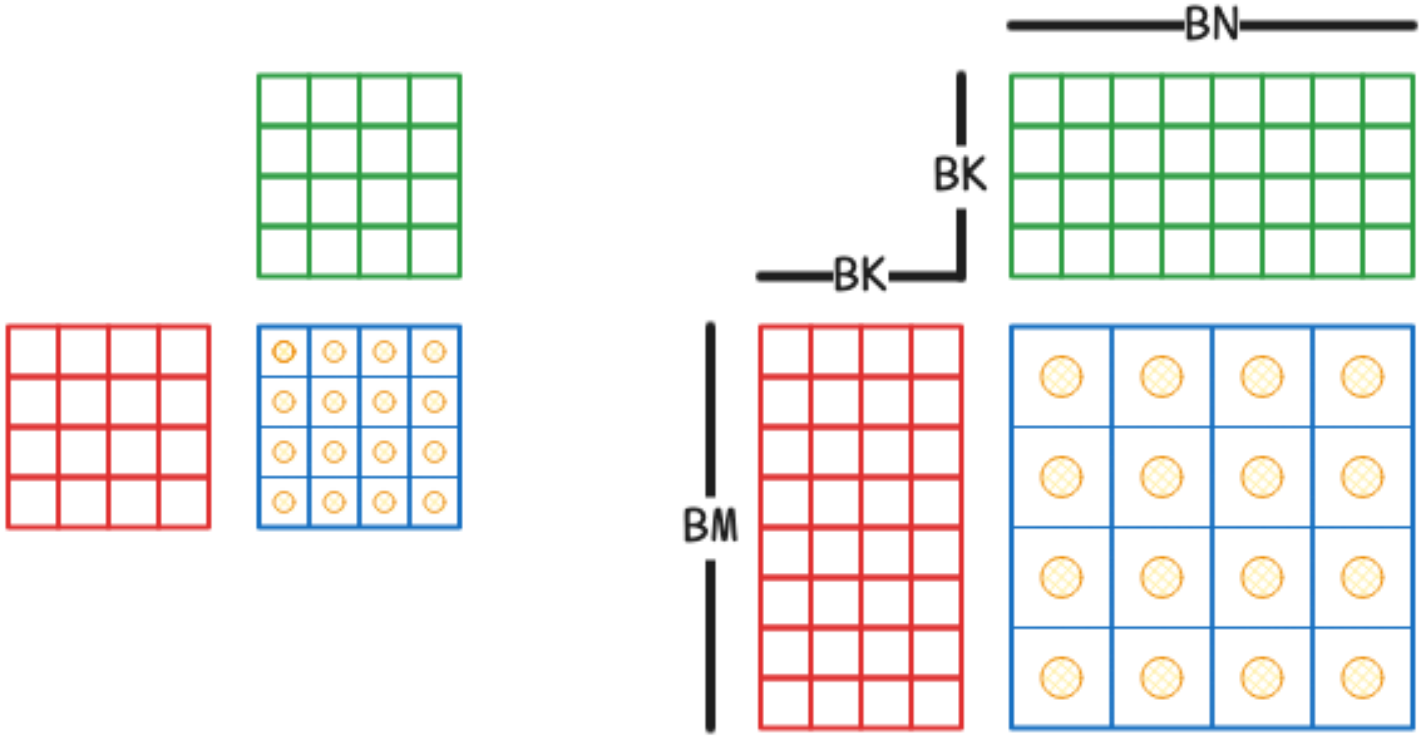
Section Overview

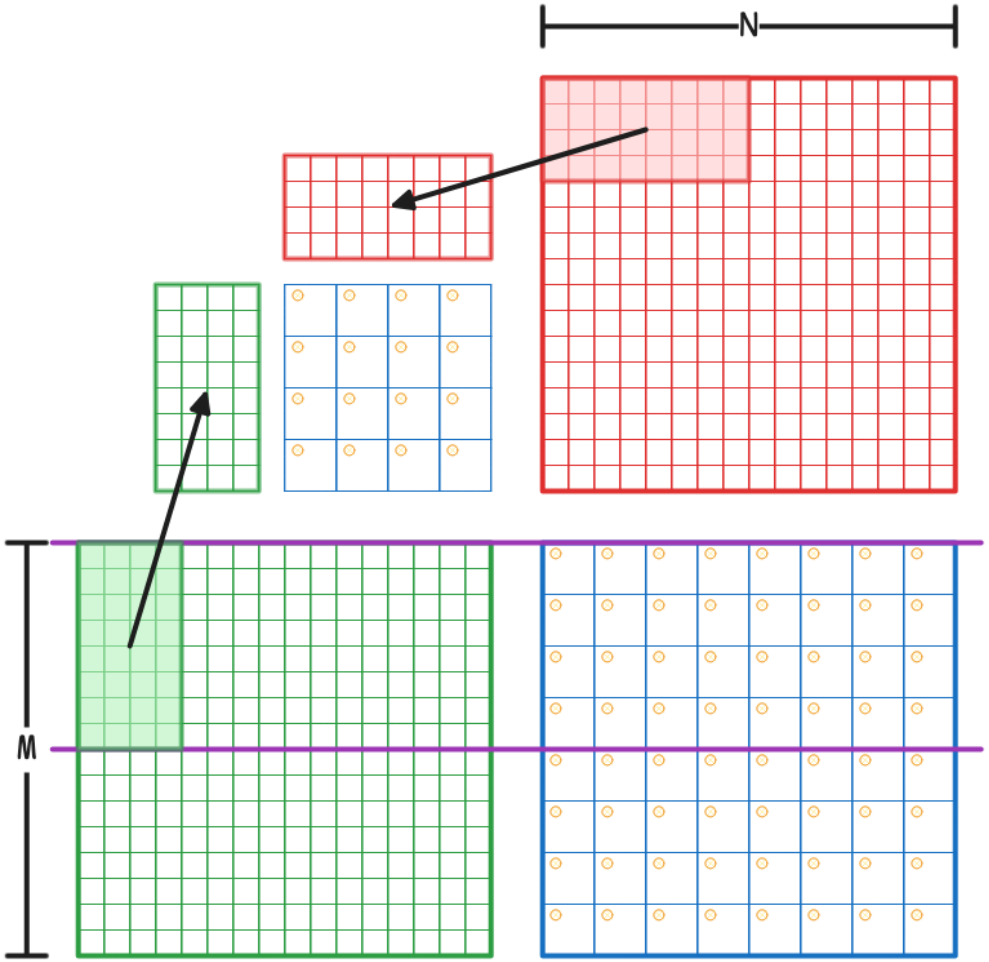
- Applying the Roofline Model: Block Tiling and Thread Coarsening

Thread coarsening: Increasing AI

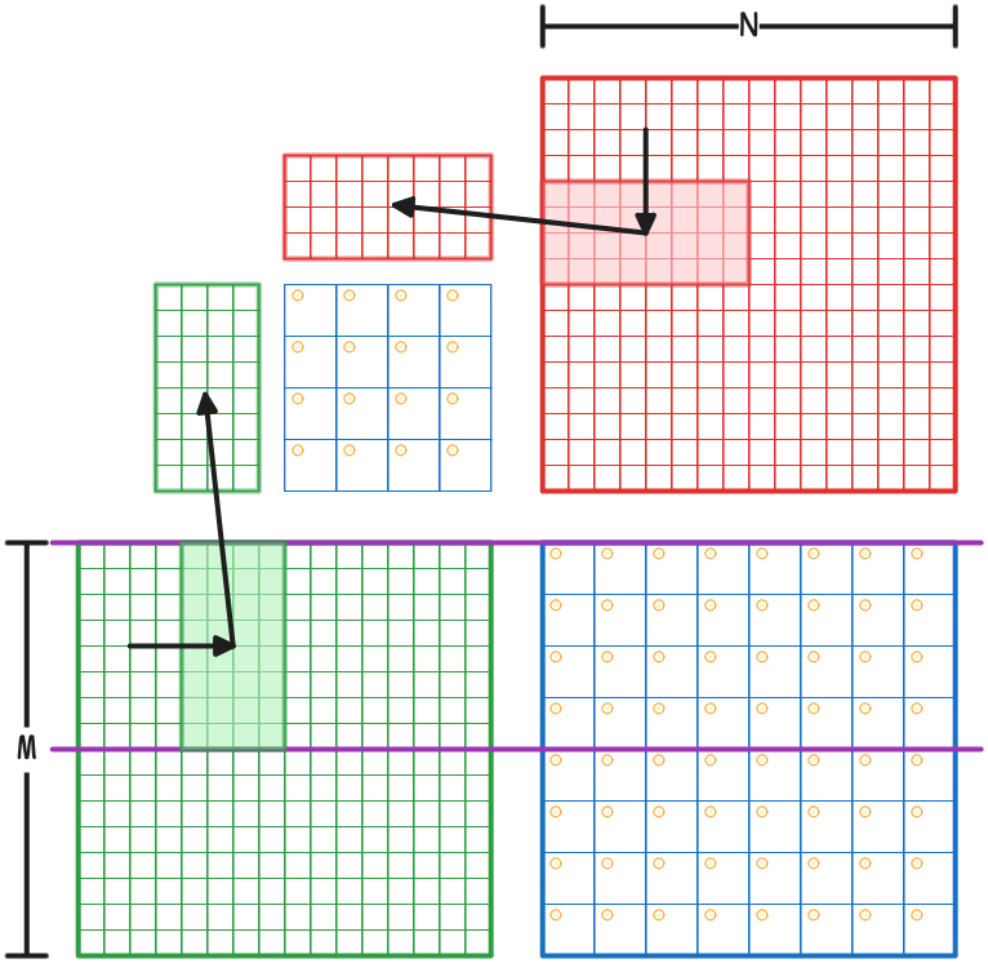
- Basic tiling assigns 1 element of the output to each thread.
- We can *coarsen* each thread, and assign a $TM \times TN$ portion of the output to each thread.
- We load larger chunks into SMEM, increasing data reuse and AI.
- One loaded A value can update several columns; one loaded B value can update several rows.



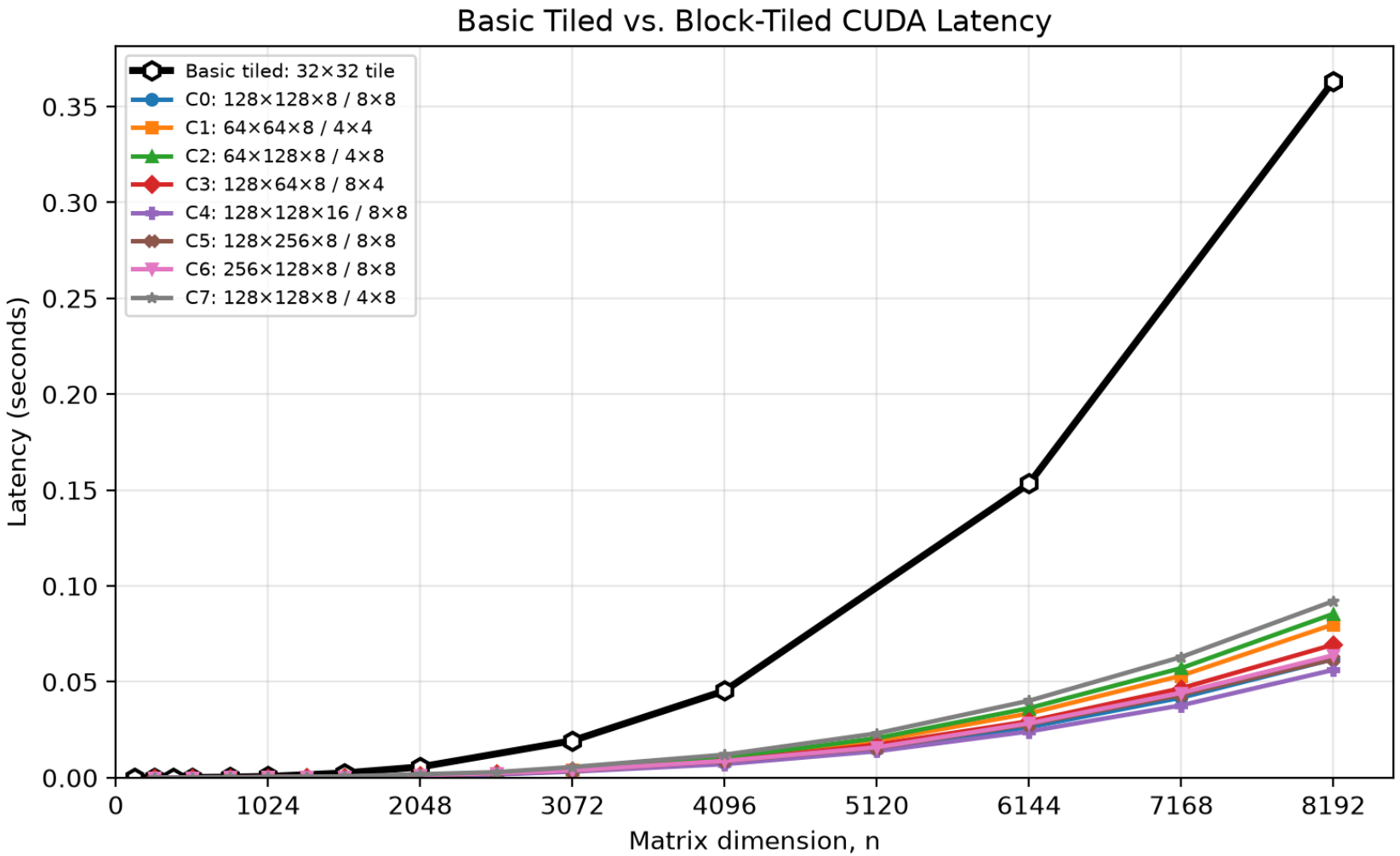
Coarsened SGEMM visualization



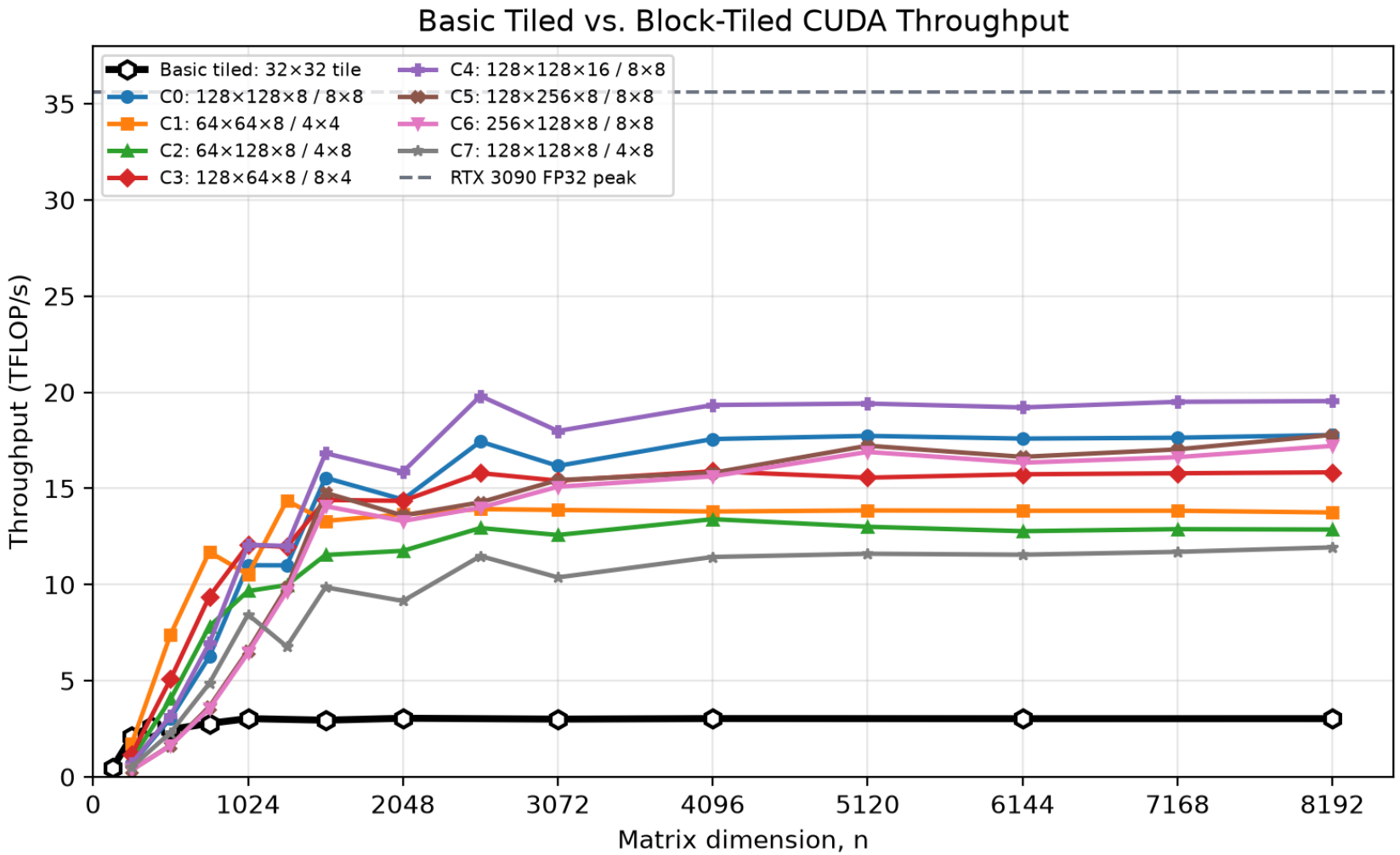
Coarsened SGEMM visualization



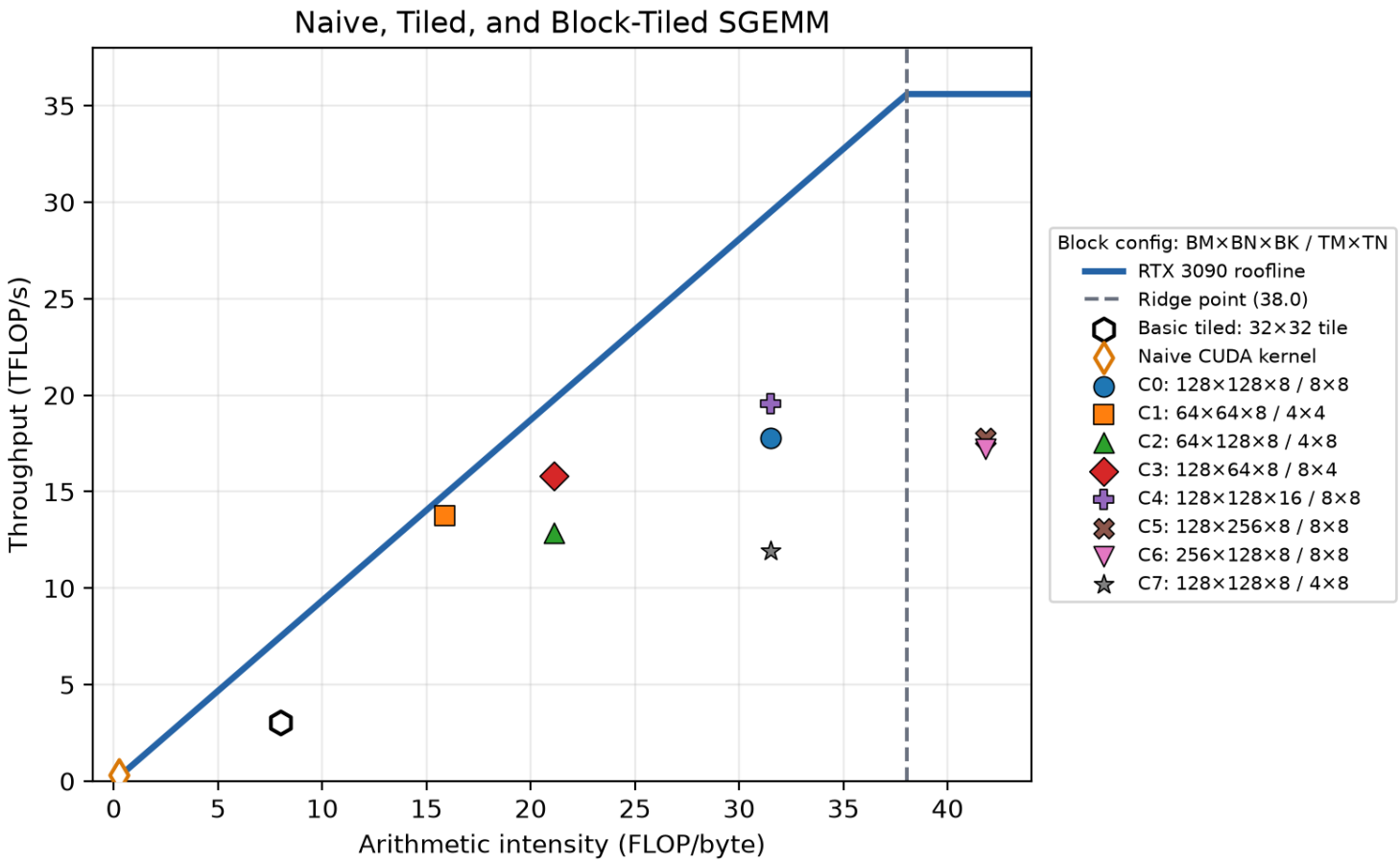
Basic tiled vs. block-tiled latency



Block tiling raises useful throughput



Tiling configurations on the roofline



Reminder that the Roofline Model only tells us the upper bound!

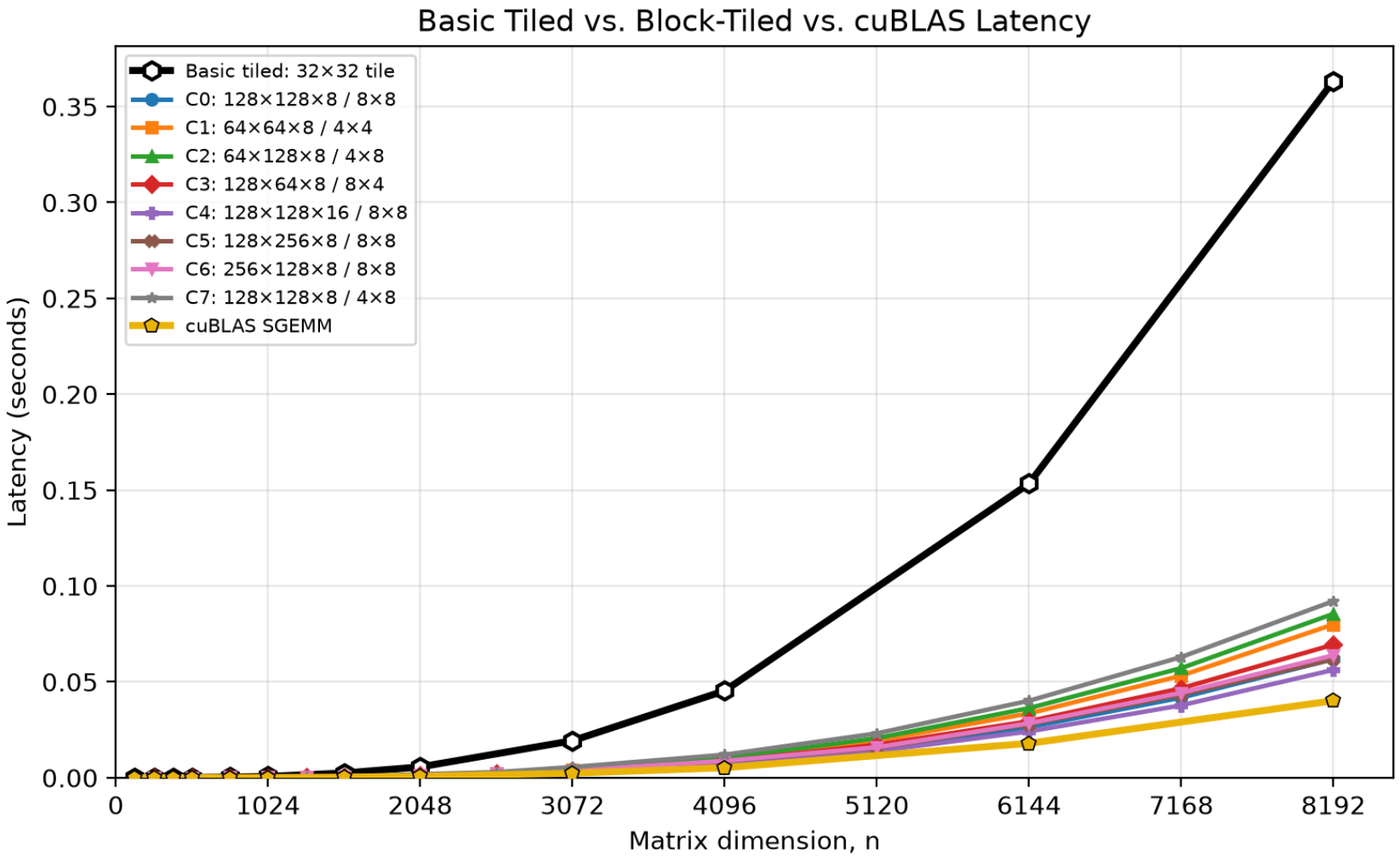
We have only scratched the surface: cuBLAS

- cuBLAS is NVIDIA's architecture-tuned linear algebra library and the practical performance target for our SGEMM kernels.
- At $n = 8192$, cuBLAS reaches 27.38 TFLOP/s, while our fastest configuration (C4) reaches 19.54 TFLOP/s, or 71.4% of cuBLAS's throughput.
- cuBLAS completes the multiplication in 40.2 ms versus 56.3 ms for C4, about 1.40× faster.

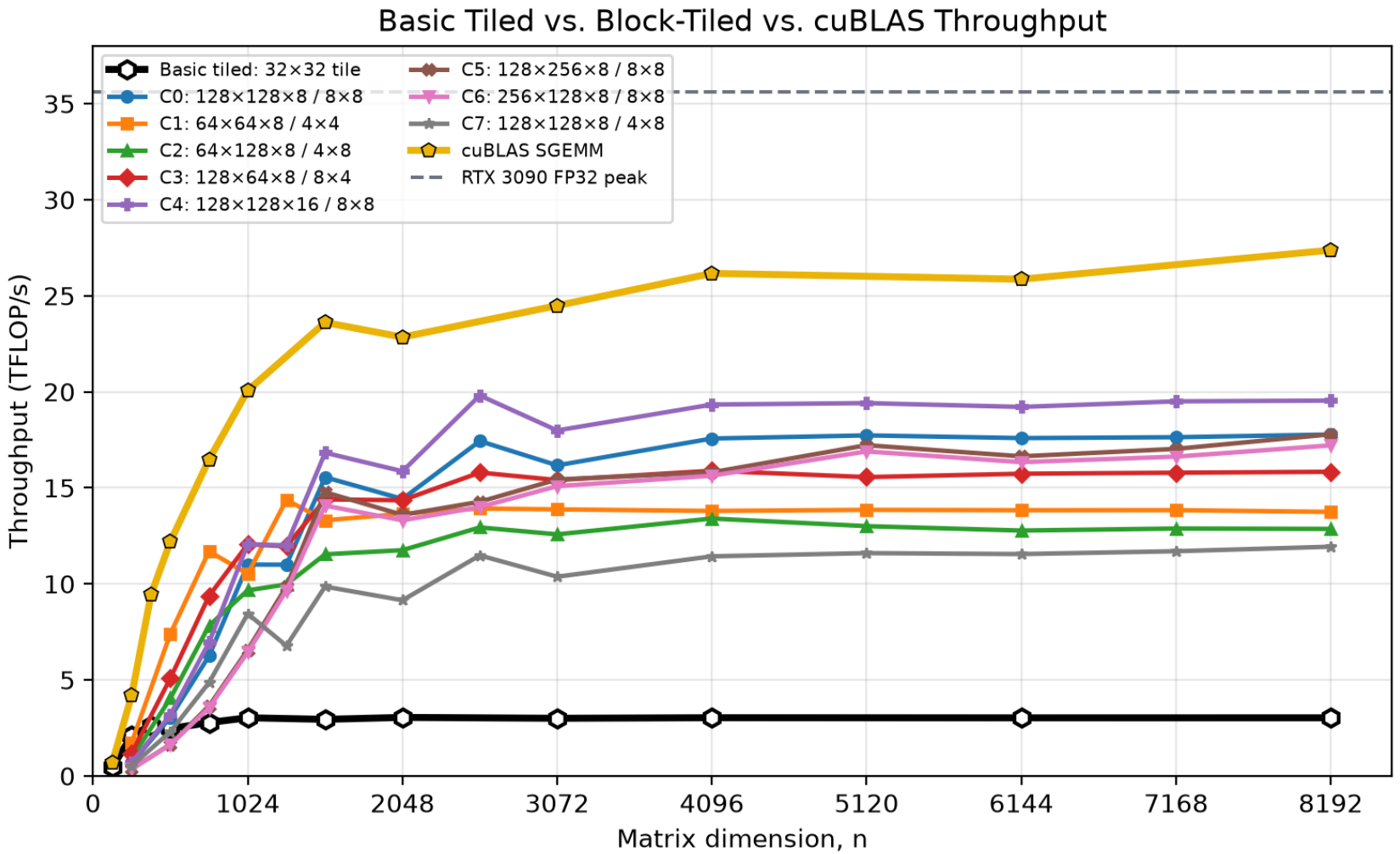
The remaining gap

Production kernels combine deeper register and warp tiling, vectorized memory access, pipelined loads, and architecture-specific instruction scheduling.

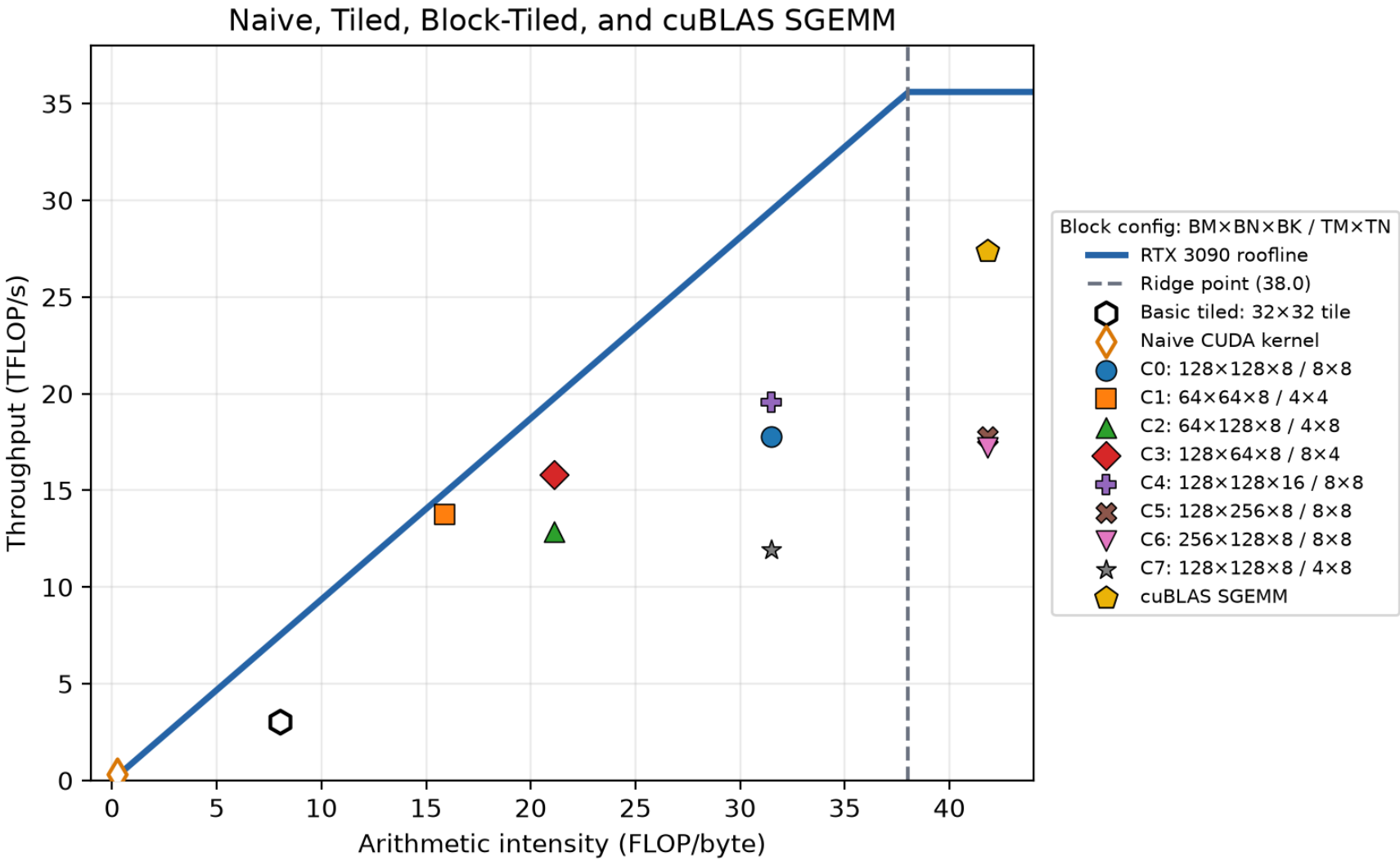
Basic tiled vs. block-tiled latency vs. cuBLAS



cuBLAS remains the performance target



cuBLAS on the roofline



cuBLAS arithmetic intensity is estimated from the profiled 256×128 output-tile shape.

We've come a long way

